Hormonal, Medical, and Nonsurgical Aspects of Gender Affirmation

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KEYWORDS

• Transgender • Gender affirmation • Cultural competency • Hormonal therapy

KEY POINTS

- Transgender people have health needs that are unique and individual and experience significant health disparities.
- Creating a safe and welcoming clinical environment involves aspects of physical environment, interpersonal communication, customer service, the electronic health record, and institutional support.
- Gender-affirming treatment with medications and/or surgery should be individualized.

INTRODUCTION

Although the acronym LGBTQ is often used as a catchall label for sexual and gender minorities, the groups included under the umbrella of that acronym—lesbian, gay, bisexual, transgender, queer/questioning—have unique and individual health needs. In this section, the authors explore health care needs and issues that are unique to transgender people.

One of the cornerstones of providing competent care to transgender people is a thorough understanding of common terminology used in conversations about sex and gender. Even physicians often conflate these 2 concepts. The following are the high-yield terms and concepts:

- *Biological sex* is composed of genotype and phenotype (chromosomes, genitalia, etc.). Although most often we think of sex as a binary (male and female), there actually is a spectrum of biological sex (XX, XY, XO, XXY, XYY, mosaicism, etc).
- *Gender*, on the other hand, is the set of roles, activities, expectations, and behaviors assigned to women and men by society. Gender is defined by culture and historical period and is thus not immutable.
- *Gender identity* is our innermost sense of whether we are male, female, both, or neither. This can change over time and can only be self-identified by an individual.
- *Transgender* is a term that has multiple usages, some broad and some narrower. Broadly, "transgender" can describe anyone whose gender identity or expression falls outside of stereotypical "norms." More narrowly and more commonly, it is used to describe someone whose gender identity differs from their assigned sex at birth.

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- *Sexual orientation* refers to romantic attraction. Our sexual orientation and our gender identity are separate, distinct parts of our overall identity, and may evolve or change over time. Being transgender does not imply any specific sexual orientation. Transgender people may identify as straight, gay, bisexual, or some other sexuality.
- *Gender dysphoria* is the term used in the Diagnostic and Statistical Manual of Mental Disorders (DSM) to describe the distress that is caused by the incongruence between an individual's gender identity and their sex assigned at birth. Not all transgender people experience gender dysphoria. The DSM clearly states that gender nonconformity itself is not a mental disorder. The critical element of gender dysphoria is the presence of clinically significant distress.[1]
- *Gender affirmation/transition* is the process of recognizing and accepting one's gender identity and then taking steps to express that gender identity. The primary foundation of transition-related care is individualization of care. There is no one right way to approach gender affirmation. The ultimate goal is increased congruence between internal gender identity and external expression and increased wellbeing. Not all transgender people want to undergo or are able to undergo medical or surgical therapy.

Table 1 lists some common terms and definitions.

Table 1
Common terms and definitions

Term	Definition
Sex (n.)	Phenotypic and/or physiologic characteristics used to assign sex at birth (eg, chromosomes, hormones, and internal and external natal genitalia). Often referred to as "assigned sex at birth."
Sexual orientation (n.)	Term that refers to being romantically or sexually attracted to people of a specific gender (eg, a lesbian or gay man is someone who has attraction to a person of the same gender).
Gender (n.)	The set of roles, activities, expectations, and behaviors assigned to women and men by society, defined by culture and historical period.
Gender identity (n.)	One's innermost sense of self as male, female, both, or neither.
Gender expression (n.)	One's external expression of gender (eg, clothing, speech, mannerisms), which are conventionally considered masculine or feminine.
Transgender, gender-expansive, gender nonconforming (adj.)	Sometimes used as umbrella terms to describe anyone whose identity or behavior falls outside of stereotypical gender norms. More narrowly, refers to a person whose gender identity does not match their assigned birth sex.

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INCIDENCE/PREVALENCE

There are several challenges to collecting accurate data on the size of the transgender population. Examples of barriers include inconsistent collection of sexual orientation and gender identity (SOGI) data both at the clinic level and at the level of the national census, as well as reliance on data from gender-specialty centers, which vastly underestimates the transgender population because only a small percentage of gender-expansive patients will seek care at these centers. More recent studies using geographic or demographic sampling techniques have reported population-based estimates ranging from 0.33% to 1.6%, which are 10- to 100-fold higher than those reported in earlier studies.[2]

HEALTH DISPARITIES

Although there have been recent strides in increased awareness and acceptance, transgender people continue to experience significant

Table 1 (*continued*)

Term	Definition
Nonbinary, gender queer (n.)	A gender identity or expression that is not strictly male or female.
Gender affirmation/ gender transition (n.)	The time period during which a person recognizes, accepts, and begins to live according to their gender identity, rather than the gender they were thought to be at birth. Each person's process is different and might include social, medical, or surgical components. (Older terms such as "sex-reassignment," "sex-change," or "cross-sex hormone therapy" can sometimes be perceived as pejorative and are falling out of usage.)
Cis-gender (adj.)	Refers to a person whose gender identity aligns with their assigned birth sex (ie, a nontransgender person).

minority stress and health disparities.[3,4] According to the most recent US Transgender Survey[4]

- When compared with the general population, transgender people experience higher rates of homelessness, poverty, psychological distress, suicidality, human immunodeficiency virus, bullying/mistreatment in school, and intimate partner violence.
- Transgender people of color experience even more profound and stark discrimination and disparities compared with white transgender people and the general US population.
- Thirty-three percent of respondents who saw a health care provider in the last year reported at least one negative experience related to gender identity (eg, refusal of treatment, verbal harassment, physical or sexual assault, needing to teach health care provider about how to properly care for transgender people).
- Twenty-three percent of respondents did not see a doctor when they needed to because of fear of mistreatment.

Transgender people still struggle to get access to gender-affirming treatments and surgical procedures. Again, from the US Transgender Survey[4]

- Fifty-five percent of respondents who had sought coverage for transition-related surgery and 25% who had sought coverage for hormones in the past year were denied.[4]
- Twenty-one percent were covered by insurance for surgery but had no surgical providers in their network.[4]
- There is often a large gap between the percentage of transgender people who would like to have a gender-affirming surgery and the percentage of those who are actually able to access this care (**Figs. 1** and **2**).

There are an increasing number of studies that demonstrate that gender-affirming surgeries improve mental and physical health of transgender people.[5] The American Medical Association's 2016 policy recognizes that "medical and surgical treatments for gender dysphoria, as determined by shared decision making between the patient and physician, are medically necessary as outlined by generally-accepted standards of medical and surgical practice."[6] Multiple other professional organizations, including the World Professional Association for Transgender Health (WPATH), the American Psychological Association, the American Academy of Family Physicians, and the American College of Obstetricians and Gynecologists have issued similar statements. The idea that medical and surgical interventions are cosmetic rather than medically necessary is out of line with current medical recommendations.[5,7]

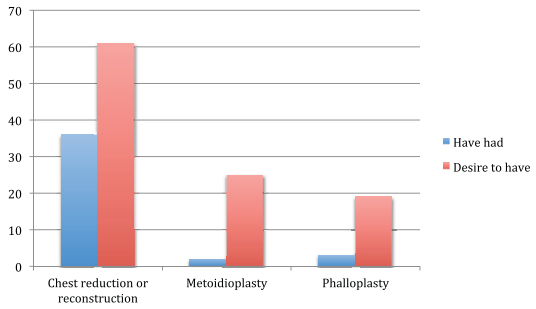

Fig. 1. Surgeries among transgender men. (*Data from* James SE, Herman JL, Rankin S, et al. The report of the 2015 U.S. transgender survey. Washington, DC: The National Center for Transgender Equality; 2016.)

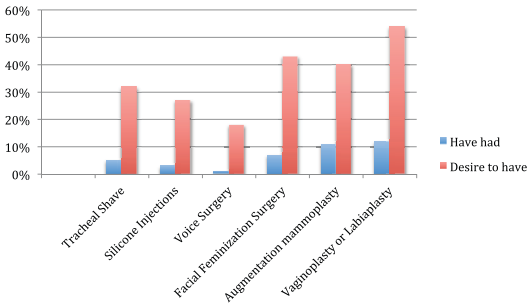

Fig. 2. Surgeries among transgender women. (*Data from* James SE, Herman JL, Rankin S, et al. The report of the 2015 U.S. transgender survey. Washington, DC: The National Center for Transgender Equality; 2016.)

These disparities and access gaps, along with the lack of in-network surgical providers who provide gender-affirming procedures, represent an opportunity for plastic surgeons to help improve access to this medically necessary care and reduce health disparities.

CREATING A WELCOMING ENVIRONMENT

The physical space of a clinic or hospital provides an important first impression for gender-expansive people seeking care. Often, patients will look for clues that a health care facility is a safe place for them to seek care.[8,9]

There are several improvements that can be implemented in the clinic or hospital environment that can create a culture of safety and welcome for transgender people:

- Window decals or stickers of the gay pride flag (rainbow) or transgender pride flag (5 horizontal stripes: light blue, pink, white, pink, and light blue) are signs of welcome to the LGBTQ community that, if placed on the front door or window, can be seen by patients as they first enter the clinic.
- Waiting room: display posters or artwork that represent LGBTQ people, have magazines or pamphlets that focus on LGBTQ issues available.
- Simply using the restroom can be a stressful and fraught experience for transgender patients. Making sure that there is a clearly marked single occupancy or gender-neutral restroom in the clinic/hospital that transgender people can access safely is an important aspect of creating a welcoming space.

VERBAL AND WRITTEN COMMUNICATION

Treating transgender patients with respect aligns with core principles of customer service and begins with a patient's first interaction with the registration desk. As with any patient encounter, maintaining privacy and avoiding collecting sensitive information in public areas are essential aspects of providing clinical care.[10]

Pronouns are words that we use in discourse to refer to others, such as she/her/hers and he/him/his. Some gender-expansive people may use gender-neutral pronouns such as they/them/their or ze/zir/zirself (eg, "Did you see the scarf they were wearing? They knitted it themselves." or "Did you see the scarf ze is wearing? Ze knitted it zirself."). Although it may seem grammatically incorrect or feel awkward at first, using correct pronouns is an essential part of providing appropriate care to gender-expansive patients because it validates gender identity and creates a therapeutic relationship. Using the wrong name or pronouns is known as "misgendering," and it is not only disrespectful and invalidating but can also lead to patients feeling unsafe in the clinical environment and avoiding care.[11]

Table 2 gives some examples of certain phrases to include (or avoid) during conversations with patients who are transgender.

Having intake forms that have LGBTQ-inclusive options and/or open-ended questions with blank spaces for patients to provide answers will increase patient comfort and inclusivity. Multiple organizations including the US Centers for Disease Control have endorsed a 2-step method for gathering gender identity data. The 2-question method uses multiple-choice questions and allows patients to self-report gender identity (ie, male, female, transgender, gender queer, additional gender identity, decline to state) as well as the sex assigned at birth (ie, male, female, intersex, decline to state). The 2-question approach has been shown to be acceptable to patients and increase response rates on questionnaires.[12]

THE ELECTRONIC HEALTH RECORD

In 2015, the Centers for Medicare and Medicaid Services and the Office of the National Coordinator for Health Information Technology ruled that in order to be certified for Stage 3 Meaningful Use, electronic health record (EHR) software must include sexual orientation and gender identity fields.[13]

Having accurate and complete SOGI data in the EHR is essential for fostering a clinical environment that is not stigmatizing or harmful for transgender people, for creating an accurate record of which organs a patient has so that their treating clinician can make accurate screening and treatment decisions, and also for building robust data sets for ongoing clinical research to assess health

Table 2
How to respond to common scenarios

Scenario	Language to Use	Language to Avoid
Patient arrives at registration desk	"How may I help you today?"	"How may I help you, sir/ma'am?"
Discrepancy between name given by patient and name on medical record	"Is it possible your chart is under a different name?"	"What's your real name?"
Unsure how to address a patient	"What name and pronouns do you use?"	Using name/pronouns you assume to be correct
Collecting SO/GI information	"Because many of our patients are affected by gender issues, I ask everyone these questions about sexual orientation and gender identity."	"Are you male or female?" "You look like a real woman/man! I never would have guessed!"
Staff member uses wrong name/pronouns	"I'm very sorry that I used the wrong name/pronoun."	"I just can't remember to use that name/pronoun." "You don't look male/female, so you can't blame me for using that name/pronoun."
Patient presents for services not related to gender	Address the clinical issue at hand	Inquiring about gender transition, what organs/genitals someone has, or what surgeries they have had

Adapted from Reisner S. Meeting the health care needs of transgender people. Available at: http://www.lgbthealtheducation.org/wp-content/uploads/Sari-slides_final1.pdf. Accessed May 30, 2018; with permission.

disparities and interventions that can benefit the LGBTQ population.[13]

Transgender people have unique needs in regard to how demographic information and medical information are recorded and displayed in the EHR. Demographic information includes data such as name, SOGI information, and pronouns. Pertinent medical information includes medications used, which organs have been surgically enhanced or removed, and which organs remain. The EHR can often obscure this information by prepopulating note templates based on the sex noted in the chart or by not clearly guiding users to use appropriate names and pronouns. The WPATH EMR Working Group has published recommendations for how to use the EHR in a way that fosters patient comfort as well as accurate medical information for the treating clinician.[14] Key recommendations on EHR use from WPATH are as follows:

- The system should include preferred name, gender identity, and preferred pronouns in demographic data.
- Provide a way to document a patient's medical history and current anatomy inventory, which is uncoupled from the demographic information about gender identity, assigned sex, or pronouns.
- The process of transitioning from one name, anatomic inventory, and/or sex should be smooth.
- The EHR should include an easily recognizable alert to notify users of a patient's proper name and pronouns (eg, a flag in the banner at the top of the screen).

INSTITUTIONAL SUPPORT

In addition to individual interactions that foster a welcoming environment, there are several institution-level interventions that play a crucial role in creating a positive and safe experience for transgender patients[8–10]:

- Nondiscrimination statements: create, post, and enforce a nondiscrimination policy that includes "gender identity and expression." Examples of such policies from health care organizations can be found here: https://www.hrc.org/hei/patient-non-discrimination-sample-policies.
- Visitation policy: creating a written policy on visitation rights and medical decision-making that explicitly includes partners, children, and family of choice for LGBTQ patients.
- Training: ongoing annual training for all staff and prompt training for new staff to maintain

competency of all employees who interact with transgender patients either in person or on the phone.
- The Human Rights Campaign has created an online resource, the Healthcare Equality Index (HEI), which health care organizations may use to evaluate the inclusiveness and patient centeredness of their policies and practices. The HEI Website also houses several resources for health care systems to create transgender-affirming hospital policies and a transinclusive workplace (https://www.hrc.org/hei).

RESOURCES FOR FURTHER READING

UCSF Center of Excellence for Transgender Health
- http://www.transhealth.ucsf.edu
- Primary care protocols for practitioners caring for gender-expansive patients
- Online learning center

World Professional Association for Transgender Health
- http://www.wpath.org/
- Interdisciplinary professional and educational organization devoted to transgender health
- Mission: to promote evidence-based care, education, research, advocacy, public policy, and respect in transgender health.
- Free download of standards of care manual
- Conferences, symposia, and trainings

National LGBT Health Education Center
- https://www.lgbthealtheducation.org
- Free online learning modules and resources
- Also offers consultation to health care organizations with the goal of optimizing quality and cost-effective health care for lesbian, gay, bisexual, and transgender (LGBT) people.

Gay and Lesbian Medical Association Transgender Health Resources
- http://www.glma.org/
- Online learning modules and glossaries
- Online referral directory for patients

National Center for Transgender Equality
- http://transequality.org/
- Social justice organization dedicated to advancing the equality of transgender people through advocacy, collaboration, and empowerment

Philadelphia Trans Health Conference
- https://www.mazzonicenter.org/trans-health
- Yearly conference that includes a Professional Development track for medical providers, behavioral health professionals, and lawyers.

Centers for Disease Control
- https://www.cdc.gov/lgbthealth/transgender.htm
- Resources for health care providers and patients

Human Rights Campaign—Healthcare Equality Index
- http://www.hrc.org/hei
- Resources available on nondiscrimination and staff training, patient services and support, employee benefits and policies, and patient and community engagement and resources for Veterans Affairs Medical Facilities.

Hormone Therapy for Transgender Individuals

The overall goal of treatment of gender dysphoria is to reduce the distress caused by the discrepancy between a person's gender identity and their sex assigned at birth. Hormonal treatment of gender dysphoria primarily works to address the incongruence in secondary sex characteristics. Feminizing or masculinizing hormone therapy is a medically necessary intervention for many individuals with gender dysphoria.[15,16] Accordingly, WPATH, the Endocrine Society, and UCSF's Center of Excellence for Transgender Health have published guidelines that include recommendations on the medical management of gender dysphoria that are widely used by specialists in transgender health care.[16–18]

The 2 key components of hormonal treatment of gender dysphoria are reducing endogenous sex hormone levels as determined by the individual's sex assigned at birth and replacing endogenous sex hormones to normal levels consistent with the individual's affirmed gender.[18] The physical changes caused by gender-affirming hormone therapy are usually associated with improved mental well-being.[19,20]

Criteria for Treatment

Criteria for initiation of gender-affirming hormone therapy as proposed by WPATH include the following[16]:

1. Persistent, well-documented gender dysphoria/gender incongruence.
2. Capacity to make a fully informed decision and to consent for treatment.
3. Age of majority in a given country.
4. Mental health concerns, if present, must be reasonably well controlled.

Endocrine Society guidelines are in agreement with these criteria; in addition, they recommend that clinicians evaluate and address medical

conditions that could be exacerbated by hormone therapy before initiation of treatment.[18].

Pretreatment Counseling and Evaluation

Treatment goals

Hormone therapy and goals of treatment must be individualized. Some individuals may seek maximum feminization or masculinization, whereas others may be satisfied with minimization of their existing secondary sex characteristics.[16,21]

Risk assessment

The patient's individual risk factors for hormone-related adverse events should be evaluated, and they should also undergo appropriate screening tests for conditions affected by hormone therapy, depending on their age and risk profile.[16] The presence of other comorbid medical conditions must also be taken into account when assessing the individualized risk/benefit ratio.[16] To that end, it is recommended that transgender patients undergo a comprehensive history and physical examination, including screening for hepatic and renal dysfunction, hyperlipidemia, and diabetes before initiation of hormone therapy.[16,18,22]

Fertility considerations

It is important that transgender patients be counseled regarding options for fertility preservation before the initiation of hormone therapy.[16,18] Although there are some case reports of transgender people using their natal gametes for reproduction after long-term use of gender-affirming hormones, the long-term effects of these medications on fertility are not well-known.[18,23–25]

Feminizing Hormone Therapy

Estrogen therapy

There are several commercially available estrogen formulations available (**Table 3**). The recommended treatment is 17β-estradiol as an oral, sublingual, transdermal, or occasionally injectable preparation.[18,26] Current guidelines do not specify a preferential order in which these therapies should be chosen, primarily because data comparing them are lacking.[18,26] However, there is some evidence that oral preparations may be more thrombogenic due to "first-pass" metabolism in the liver, and therefore transdermal, injectable, and sublingual preparations that bypass this route may be advantageous in older patients or those otherwise at increased risk for thromboembolic events.[18,26,27]

Generally, use of injectable estradiol is less common and is more often used in patients who do not achieve hormonal goals despite using the maximum recommended dose of either oral or transdermal estrogen preparations.[26] Oral ethinyl estradiol in particular has been shown to carry a higher risk of venous thromboembolism, and therefore its use is not recommended.[27] Conjugated and synthetic estrogens cannot be monitored through blood tests and therefore are of limited use.[26]

Androgen suppression

Treatment with estrogen alone is usually insufficient to reduce testosterone levels to the physiologic range for cis women, so transgender women generally require adjunctive antiandrogen therapy.[18,28] The most commonly used

Table 3
Hormone regimens for transgender women

Medication	Route	Recommended Dose	Considerations
Estrogen	Oral	2–6 mg/d	Higher thromboembolic risk
	Transdermal	0.025–0.2 mg/24 h once or twice weekly	Possible skin reaction, more constant blood levels
	Parenteral	5–20 mg IM every 2 wk or 2–10 mg IM weekly	
Cyproterone acetate	Oral	25–50 mg/d	Not available in United States, possible liver toxicity
GnRH agonists	Parenteral	Leuprolide: 3.75 mg SC monthly or 11.25 mg SC every 3 mo Goserelin acetate: 3–6 mg SC monthly or 10–8 mg SC every 3 mo	Expensive
Spironolactone	Oral	100–300 mg/d	

Abbreviations: GnRH, gonadotropin-releasing hormone; IM, intramuscular; SC, subcutaneous.
Data from Refs.[18,26,30]

Table 4
Hormone regimens for transgender men

Medication	Route	Recommended Dose	Considerations
Testosterone	Transdermal		
	Patch	2.5–7.5 mg/d	Possible skin irritation
	Gel	50–100 mg/d	Mimics physiologic secretion, possible transfer to partners
	Parenteral		
	Enanthate, cypionate	100–250 mg IM (SC) every 2–4 wk or 50–100 mg IM weekly	Supraphysiologic levels after injection, with decline before next administration
	Undecanoate	1000 mg every 12 wk	More stable blood levels

Data from Refs.[18,23,26,41]

antiandrogen therapies include spironolactone, cyproterone acetate, and gonadotropin-releasing hormone (GnRH) agonists (**Table 4**). Spironolactone inhibits testosterone secretion, blocks androgen receptor binding, and may also have estrogenic activity.[18,26] Cyproterone acetate is a synthetic progestogen compound with antiandrogen properties widely used in Europe but is not available in the United States due to concerns for liver toxicity.[29] GnRH agonists are used to inhibit gonadotropin secretion and thereby induce profound suppression of testosterone secretion; they are given as long-acting injections and are highly effective, however their cost can be prohibitive.[16,29]

The 5α-reductase inhibitors are also sometimes used. These medications block conversion of testosterone to the potent androgen dihydrotestosterone. Although Endocrine Society guidelines do not specifically recommend their use because they do not reduce testosterone levels and may be associated with adverse effects,[18,29,30] other groups including UCSF and WPATH note that they may be beneficial especially for individuals unable to tolerate or with contraindications to spironolactone or for those with persistently virilized features despite complete androgen blockade.[16,31]

Other therapies

Progestins have sometimes been used for the treatment of transgender women. Reports of possibly improved breast development have been associated with their use. However, they have not been well studied as part of a gender-affirming hormone regimen, and overall there is little data to either support or dissuade their use. Because of this lack of data and concerns regarding possibly increased risk of breast cancer and cardiovascular disease (based on data in menopausal and postmenopausal nontransgender patients), Endocrine Society and WPATH guidelines do not specifically recommend their use.[16,18,26,29] However, other groups including UCSF note that the risk in transgender patients is likely minimal-to-absent.

Clinical outcomes

Initial physical changes include decreased body hair, decreased libido and spontaneous erections, decreased lean muscle mass and strength, decreased oiliness of the skin, increased weight, and redistribution of fat mass with an increase in subcutaneous fat.[18,32,33] Long-term treatment will result in atrophy of the prostate and testes.[18,33] Facial hair is also reduced with hormonal therapy; however, elimination is difficult to achieve. Some patients may choose to pursue additional measures such as electrolysis or laser treatments to achieve their desired appearance.[26,33] Breast development begins within the first 3 to 6 months of treatment and is typically at its maximum by 2 years.[29,33] Approximately 50% of transgender women will be dissatisfied with the breast growth attained from hormone therapy alone, prompting some to seek gender affirming surgeries.[32]

Monitoring

Transgender women should be monitored closely following initiation of hormone therapy. Current guidelines advise regular clinical evaluations and assessment for physical changes of feminization/masculinization and for possible adverse effects from hormone therapy, as well as laboratory monitoring of sex steroid hormone levels every 3 months during the first year of treatment and then 1 to 2 times yearly.[18]

A key issue in monitoring transgender women is to avoid supraphysiologic levels of estrogen that may lead to increased risk of adverse events. Goal hormone levels are within the physiologic cis-female range. This corresponds with serum testosterone levels less than 50 ng/dL and serum

estradiol levels that should not exceed the peak physiologic range of 100 to 200 pg/mL.[16,18] For patients on spironolactone, it is also recommended to monitor serum electrolytes.[18]

Pituitary lactotroph cells contain estrogen receptors, and thus up to 20% of patients receiving estrogen therapy may have increased prolactin levels while on gender-affirming medications.[16,18,29] The risk of development of prolactinoma is thought to be very low; however, some current guidelines advise monitoring prolactin levels at baseline, annually during the transition period, and at least every 2 years thereafter.[18] Other guidelines recommend checking prolactin levels only if symptoms of prolactinoma (visual symptoms, excessive galactorrhea, new onset headaches.[31]

Risks and adverse events

Venous thromboembolism (VTE) may be a serious complication in transgender women treated with estrogen. A 20-fold increase in venous thromboembolic events was seen in a study of Dutch transgender patients; however, this increased risk may have been related to the use of ethinyl estradiol, which carries an increased risk of VTE.[18,26,34] More recent studies have shown a lower risk of VTE.[35–37] Most recently, a large cohort study showed an increased risk of VTE and ischemic stroke with the use of gender-affirming hormones. Clot rates in trans women were about 5 times higher after two years of follow up when compared to cisgender men. Ischemic stroke risk in transgender women was approximately 10 times higher when compared to cisgender men, and about 4 times higher compared to cisgender women. Though this data suggests increased cardiovascular risks for transgender women, it is important to note that the actual number of events recorded in the study was 148 out of the 2,842 transgender women in the eight-year period of the study. While this new data should serve to underscore the importance of informed consent discussions and thoughtful consideration of which formulation of estrogen to use, it is important to note that for many transgender people, these medications are life-saving, and patients may decide that the associated risks are not high enough to dissuade them from starting gender affirming therapy.[38] Transdermal estrogen is recommended for older patients or those at increased risk of VTE due to avoidance of "first pass" metabolism and therefore reduced thrombogenicity, as discussed earlier. Note that guidelines recommend against screening for hereditary thrombophilia in most patients; however, it could be considered for those with a personal or family history of thromboembolic events.[16,18]

Overall, data regarding bone mineral density (BMD) and osteoporosis in transgender patients are limited. Estrogen has been shown to preserve BMD in patients who continue on therapy.[18,29] However, transgender patients who undergo gonadectomy and then choose to discontinue hormone therapy are clearly at increased risk for bone loss. Fracture data are not known for transgender patients, nor is it known whether to use sex assigned at birth or affirmed gender when estimating fracture risk.[18]

As described above, there is conflicting data regarding hormonal therapy in transgender women and cardiovascular risk. It is recommended that transgender women be assessed for cardiovascular risk factors with lipid panels and diabetes screening according to established guidelines.[18]

Masculinizing Hormone Therapy

Androgens

There are several androgen preparations available with demonstrated clinical efficacy (see **Fig. 1**). Testosterone is most often administered either parenterally or transdermally. Parenteral formulations include testosterone enanthate or cypionate, which can be injected intramuscularly every 1 to 2 weeks, and testosterone undecanoate, which may be given intramuscularly at longer intervals up to 10 to 12 weeks.[18,23,39]

Transdermal testosterone may be administered via gel or patch. Transdermal preparations may be associated with risk of skin irritation and risk of testosterone gel transfer to others. Short-term studies have shown no difference between parenteral or transdermal preparations with regard to body composition, metabolic parameters, safety, compliance, or satisfaction.[23,40]

Clinical outcomes

Testosterone therapy in transgender men results in clinical effects of virilization. During the first 1 to 6 months of treatment, expected changes include increased muscle mass, redistribution of fat mass, increased sexual desire, increased oiliness of skin, increased facial and body hair, and cessation of menses. Changes that may appear later (but still within the first year of treatment) include deepening of the voice, clitoromegaly, and sometimes male pattern hair loss.[18,23,32] It is important to note that height and bone structure are not altered by testosterone therapy.[23] In addition, testosterone therapy may decrease glandular activity of the breast but will not decrease breast size.[32]

Acne may be a significant undesired effect of testosterone therapy. The prevalence and severity of these lesions usually peaks by 6 months and tends to resolve in transgender men who continue therapy long-term.[23,41]

Monitoring

As with transgender women, transgender men should be monitored closely following initiation of treatment with regular clinical evaluations including assessment for physical changes of masculinization and possible adverse effects of hormone therapy, as well as regular laboratory monitoring.[16,18]

A major goal of monitoring treatment is to achieve testosterone levels in the normal cis-male range, which is assay dependent but typically between 320 and 1000 ng/dL.[18,28] Supraphysiologic levels of testosterone should be avoided because this may increase the risk of adverse reactions.

It is recommended that testosterone levels be monitored every 3 months until levels are in the normal physiologic cis-male range or for the first year of treatment (whichever is longer) and then 1 to 2 times annually thereafter. Current guidelines also recommend that hematocrit be monitored at baseline, every 3 months during first year of treatment and annually thereafter. When interpreting hematologic laboratory results, the "male" range should be used. Weight, blood pressure, and lipids should also be monitored at regular intervals.[16,18,39]

Risks and adverse events

A major concern with testosterone therapy has been the possibility of cardiovascular adverse events. Testosterone therapy in transgender men has been shown to result in a more atherogenic lipid profile with increased low-density lipoprotein and triglycerides and decreased high-density lipoprotein.[42] However, 2 meta-analyses did not show any increased incidence of cardiovascular events in patients on testosterone therapy,[43,44] and a long-term study in transgender men found no increased risk of cardiovascular mortality.[34] Current guidelines recommend that clinicians should screen for and manage cardiovascular risk factors as they emerge according to established screening guidelines.[16,18]

Testosterone therapy has been found to prevent bone loss in transgender men who have undergone gonadectomy.[45] However, some transgender men on hormone therapy do not maintain bone mass; this is thought to be related to low testosterone dosing and nonsuppressed luteinizing hormone concentrations, demonstrating the importance of adequate dosing of testosterone for maintenance of bone mass.[46] As with transgender women, transgender men who undergo gonadectomy and then stop testosterone therapy are at high risk for bone loss.[18]

Perioperative Concerns Related to Hormone Therapy

Timing of gender-affirming surgery

Current guidelines recommend that transgender patients use gender-affirming hormones for 1 year before gender-affirming surgery involving genitals or gonads.[16,18]

For transgender women, it is recommended that breast augmentation surgery be delayed until the patient has completed at least 2 years of estrogen therapy because breast development may continue throughout this period. However, hormone therapy is not a prerequisite for breast or chest surgery for transgender men or women.[16,18,22]

Risk of venous thromboembolism

Perioperative venous thromboembolism may be a risk in transgender women treated with estrogen, and surgical procedures and associated immobilization confer additional risk for thromboembolic events. Data on the incidence and risk of venous thromboembolism in transgender patients treated with estrogen undergoing surgery are limited. However, one review of 60 transgender patients undergoing gender-affirming surgery after 1 year of hormonal therapy did not report any venous thromboembolic events.[47]

Given the perceived increased risks, hormone therapy is frequently held perioperatively in transgender patients. However, there is currently minimal data to guide this decision. Some investigators suggest that estrogen therapy should be discontinued between 2 and 4 weeks before gender-affirming surgery, with resumption of therapy 3 to 4 weeks following surgery, or once patients are fully mobilized again.[32,36] It is common practice to temporarily interrupt hormone therapy for longer surgical procedures such as vaginoplasties, which also require modified bed rest for up to a week postoperatively. However, hormones may sometimes be continued for outpatient procedures or those that are completed in less than 4 to 5 hours. Current guidelines suggest that the surgeon and hormone-prescribing physician collaborate in making a decision about the use of hormones before and following surgery.[18]

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